Max Shen

chihhans@usc.edu ◇ github.com/maxshen1212 ◇ linkedin.com/in/maxshen1212

Education

University of Southern California, Los Angeles, CA — *Jan 2026 – May 2027*
M.S. in Computer Science — *GPA: 4.0 / 4.0*

National Chin-Yi University of Technology, Taichung, Taiwan — *Sep 2019 – Jun 2023*
B.Eng. in Computer Science and Information Engineering

Research Experience

SLURM Lab, USC, Advisor: Prof. Daniel Seita — *Jan 2026 – Present*
Student Researcher [World Models, VLM, Perception, MuJoCo] — *Los Angeles, CA*

Project: Hallucination Detection for World-Model Policy Evaluation (Targeting RSS 2027)

- Generating CTRL-World rollouts on DROID for a 240-pair, 16-task world-model hallucination benchmark.
- Co-building VLM hallucination-detection pipelines with Qwen3-VL and Molmo2 over frame pairs, sliding-window clips, and full videos, using SAM3 masks to target task-object disappearance and shape morphing in rollouts.

Project: Tactile Real-to-Sim-to-Real for Dexterous Manipulation

- Recovered 98% of applied contact force (vs. 22% for MuJoCo's built-in sensor) by designing an elastic-pad model that simulates 127-taxel PaXini fingertip sensors on a LEAP hand in MuJoCo, following contact shape and push direction.
- Narrowing the tactile sim-to-real gap by benchmarking 5 sensor models against real recordings replayed in sim.

Smart Manufacturing Lab, NCUT, Advisor: Prof. Yu-Chi Liu — *Aug 2022 – Jul 2023*
Student Researcher [Machine Learning, Embedded Systems, C++] — *Taichung, Taiwan*

Project: Adaptive Thermal Equilibrium Control System for CNC Machines

- Achieved 0.5 μm RMSE thermal-tilt prediction by co-developing an MLP that feeds a MOPSO setpoint optimizer.
- Held spindle tilt within $\pm 1\,\mu$m (from -3 to $4\,\mu$m) via closed-loop control fed by 20 co-built RP2040 wireless sensors.

Professional Experience

Graphen – R&D Team — *May 2026 – Aug 2026*
Robotics AI Intern [Python, PyTorch, Isaac Sim, VLA, World Model, ROS2] — *Taipei, Taiwan*

- Achieved 60% real-robot success on a 40s+ long-horizon bimanual pick-and-place task by extending a single-arm platform to synchronized dual-arm control and training a Diffusion Transformer (DiT) imitation policy.
- Doubled demonstration throughput to 50 episodes/hr by developing the real-robot data collection pipeline that integrates teleoperation, MoCap, and MoCap gloves to capture bimanual and dexterous-hand demonstrations.
- Removed 200+ invalid episodes (∼20% of collected data) by designing an automated data-QA pipeline that rejects statistical-outlier trajectories, frozen-frame episodes, timestamp errors, and misaligned action–camera streams.
- Collected 400 simulated episodes for sim-to-real Vision-Language-Action (VLA) post-training of GR00T via Isaac Lab with domain randomization, and prototyped an end-to-end NVIDIA Cosmos world-model data pipeline.

DoubleSpot — *Sep 2025 – Dec 2025*
Software Engineer Intern [TypeScript, Docker, AWS, Terraform, CI/CD] — *Seattle, WA*

- Cut development cycles from two months to one week and new-engineer onboarding to two days by proposing a full re-architecture to the CTO and designing the new TypeScript, React, and Node.js stack end to end.
- Onboarded 100+ client organizations onto a multi-tenant AWS platform (ECS, ALB, RDS, S3) designed with Terraform, and replaced fully manual releases with automated, reproducible Dockerized CI/CD deployments.

Yois — *Aug 2023 – Jul 2025*
Software Engineer [TypeScript, Java, SQL, CI/CD] — *Taipei, Taiwan*

- Developed and maintained mobile and web apps, integrating backend REST APIs and writing automated API tests.

Skills

Programming Languages: Python, C++, TypeScript, SQL
Robot Learning: Reinforcement Learning, Imitation Learning, Diffusion Policy, VLA, VLM, World Models, Tactile
Tools: PyTorch, Isaac Sim/Lab, MuJoCo, ROS2, LeRobot, Cosmos, OpenCV, Docker, Linux, Git, AWS
Certifications: NVIDIA Isaac for Accelerated Robotics (Jun 2026)